PROBLEMS ENCOUNTERED

N/A

INTRODUCTION

The purpose of this lab is to explore the applications of operational amplifiers, utilizing them as foundational building blocks for more complex circuits. More specifically, this lab demonstrates how op-amps can be configured for buffering to prevent loading effects, level shifting to adjust signal voltage ranges, and converting differential signals into single-ended outputs with improved noise performance.

DISCUSSION

1.1 Objective

The purpose of this lab is to delve into the applications of operational amplifiers. There is an endless number of applications, the possibilities are often not intuitively obvious for students seeing them for the first time.

1.2 Materials

- Laptop with LTSpice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit with TLV272

1.3 Introduction

There are several standard op amp configurations. While Lab 5 demonstrated the simple inverting and non-inverting cases, this lab expands the scope to include both a summing amplifier and a difference amplifier.

Summing Amplifier

A summing amplifier, Figure 6.1 (c), is effectively the same configuration as an inverting amplifier but has multiple inputs that allow for a weighted output voltage.

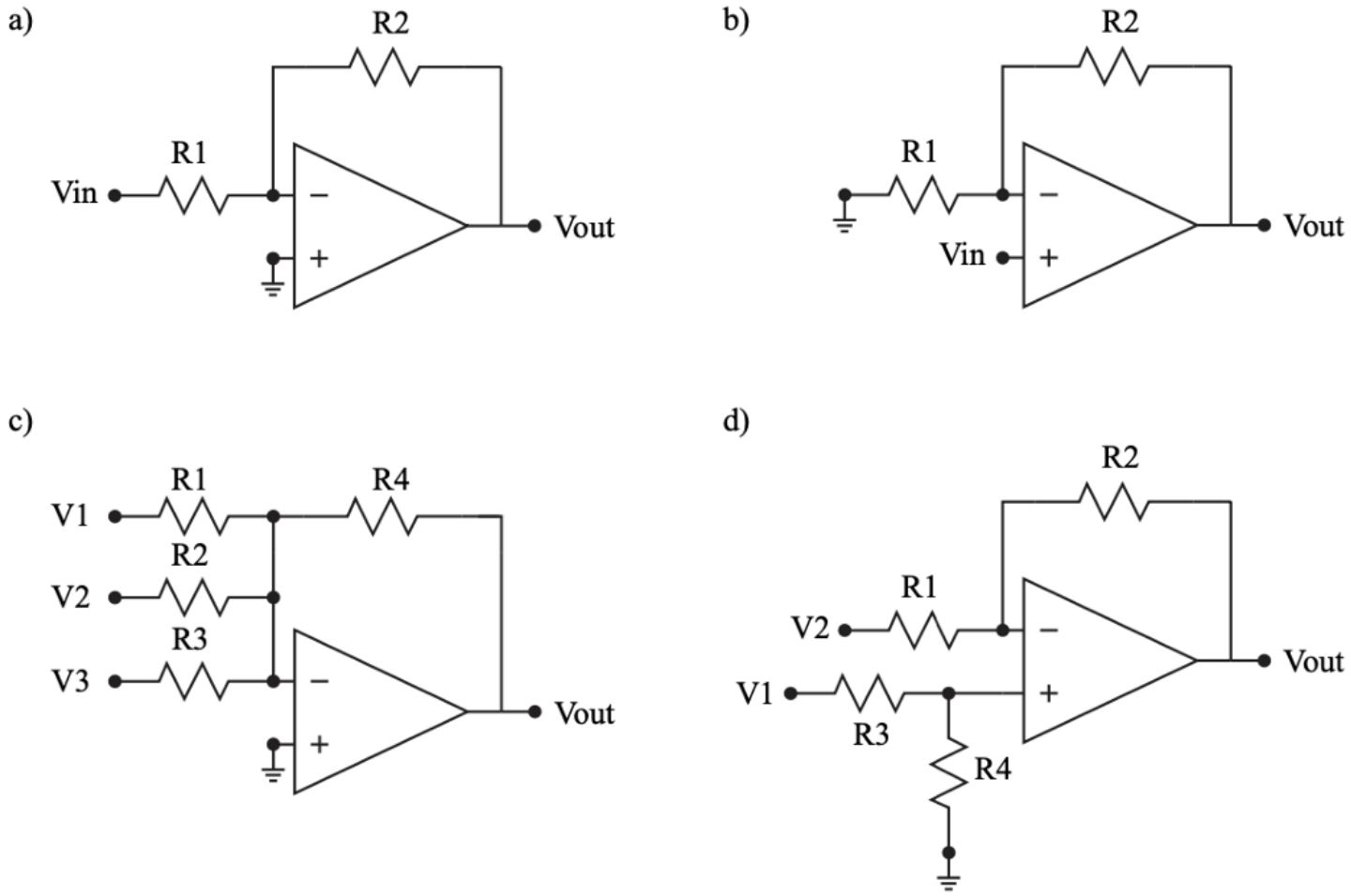


Figure 1: Different op amp configurations: inverting amplifier (a), non-inverting amplifier (b), summing amplifier (c), and difference amplifier (d).

$$V_{out} = -\left(\frac{R_4}{R_1}V_1 + \frac{R_4}{R_2}V_2 + \frac{R_4}{R_3}V_3\right) \qquad (6.1)$$

When the resistors R_1, R_2, R_3, and R_4 are equal, the output voltage is an equal combination of the input voltages. However, the resistors can be adjusted to assign different weights to the input voltages based on the application.

Difference Amplifier

Difference amplifiers, Figure 6.1 (d), takes the difference between two input voltages, with the output determined by the resistor value ratios.

$$V_{out} = V_1\left(\frac{R_2}{R_1+R_2}\right)\left(\frac{R_3+R_4}{R_3}\right) - V_2\frac{R_4}{R_3} \qquad (6.2)$$

When the resistors $R_1 = R_3$ and $R_2 = R_4$ the output voltage simplifies to:

$$V_{out} = (V_1 - V_2)\frac{R_2}{R_1} \qquad (6.2)$$

1.4 Big Picture

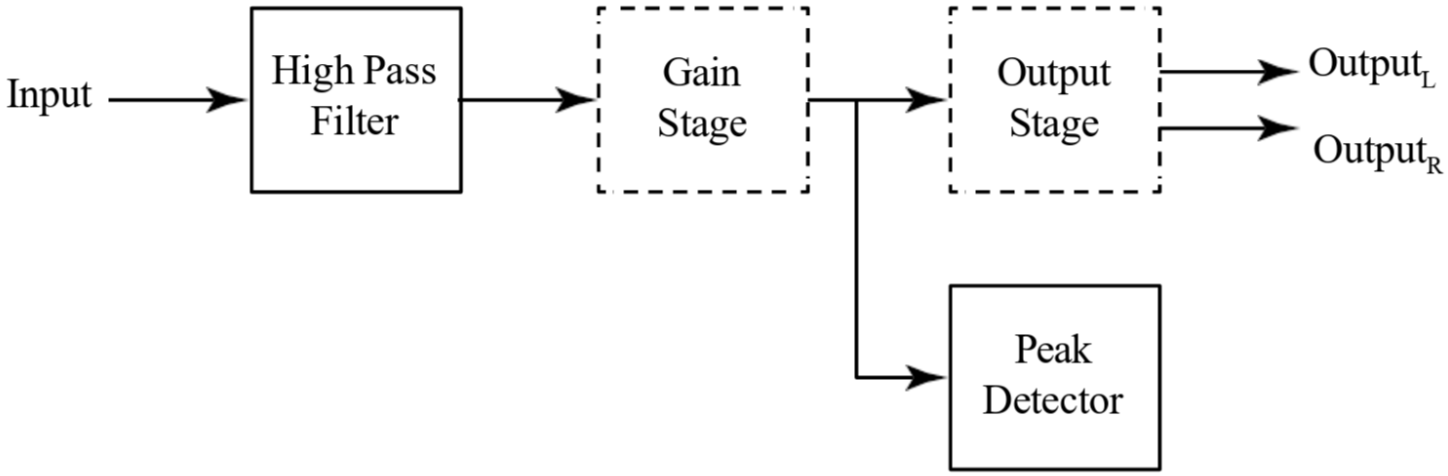


Figure 2: Big picture with emphasis on the gain and output stages.

This lab once again focuses on op amps, using them as building blocks to create more complicated circuits. The final project is about as complicated as the final circuit in this lab.

1.5 Pre-Lab Requirements

Complete the following before coming to lab.

1.5.1 Buffering

Buffering is a key op amp application that addresses loading effects, which resistive circuits cannot easily solve. Op amp buffers provide high input impedance and low output impedance, preserving signal integrity across various applications.

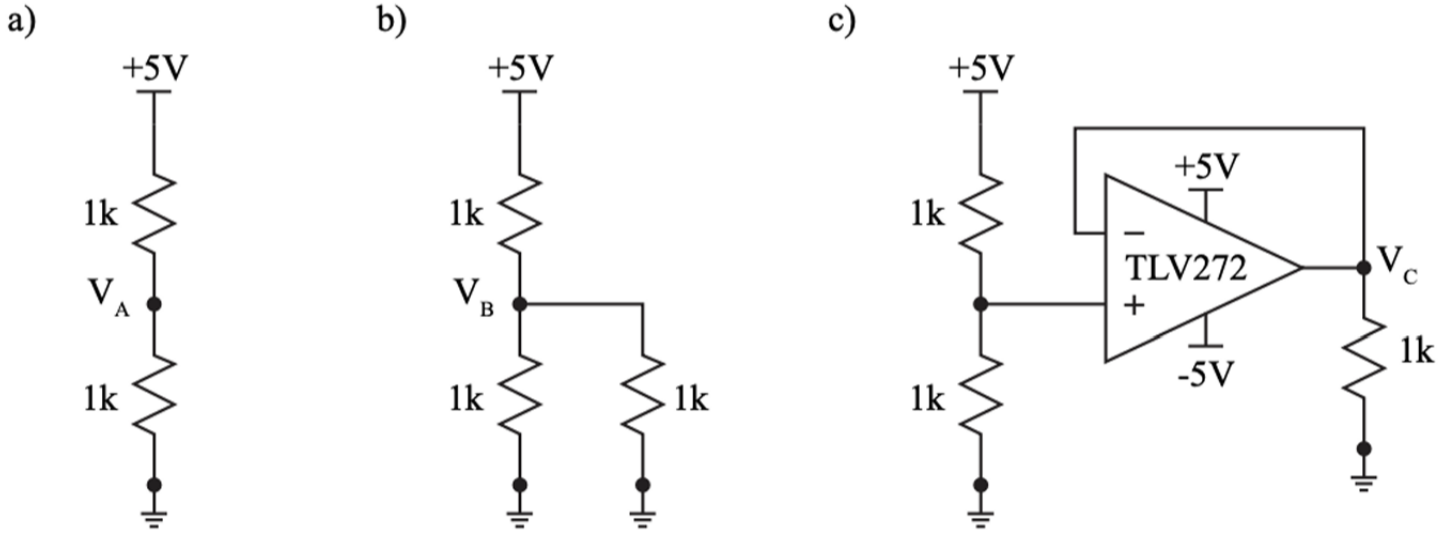
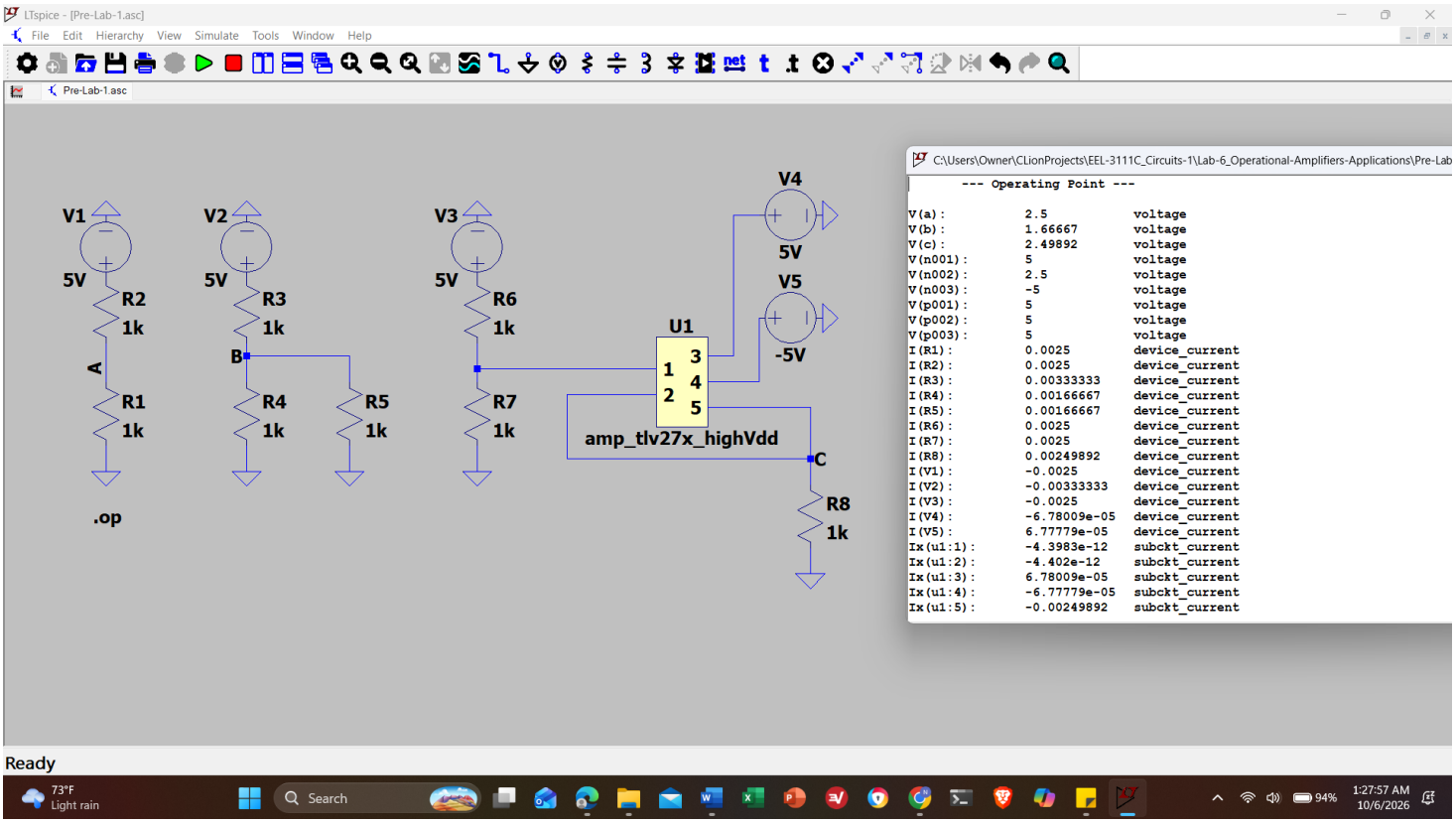


Figure 3: A simple voltage divider (a), voltage divider with a load (b), and a voltage divider with a voltage follower between the output and the load (c).

1. Simulate all three circuits in Figure 6.3 using a DC operating point (.op). Keep all the circuits on the same schematic and table the voltages V_A, V_B, and V_C. Table your values and include images of your circuits.



Screenshot 1: All three circuits in Figure 6.3 using a DC operating point (.op).

Node	Expected Voltage (V)
V_A	2.50000
V_B	1.66667
V_C	2.49892

Table 1: Tabled Values of Figure 6.3 LTspice circuits

2. Discuss why two of the voltages are approximately 2.5 V but one is not.

   The Voltages V_A and V_C are about 2.5V because V_A is a perfect unloaded divider, and V_C is buffered by the op-amp, which draws negligible current and prevents the 1k load from pulling down the voltage. V_B is 1.67 V because the 1kΩ load is parallel with the bottom 1kΩ resistor of the divider, creating an equivalent resistance of 500Ω and so it draws more current from the top half of the divider.

1.5.2 Level Shifting

Level shifting in electronic circuits is the process of changing the supply rails from one range to another. For example, when preparing a signal for analog to digital conversion, the signal

may be bi-polar (+/-) and must be converted to a single voltage rail. So, instead of a sine wave that swings to +2.5V and -2.5V, 2.5V is added so that the sine wave swings from +5V to 0V with the same amplitude.

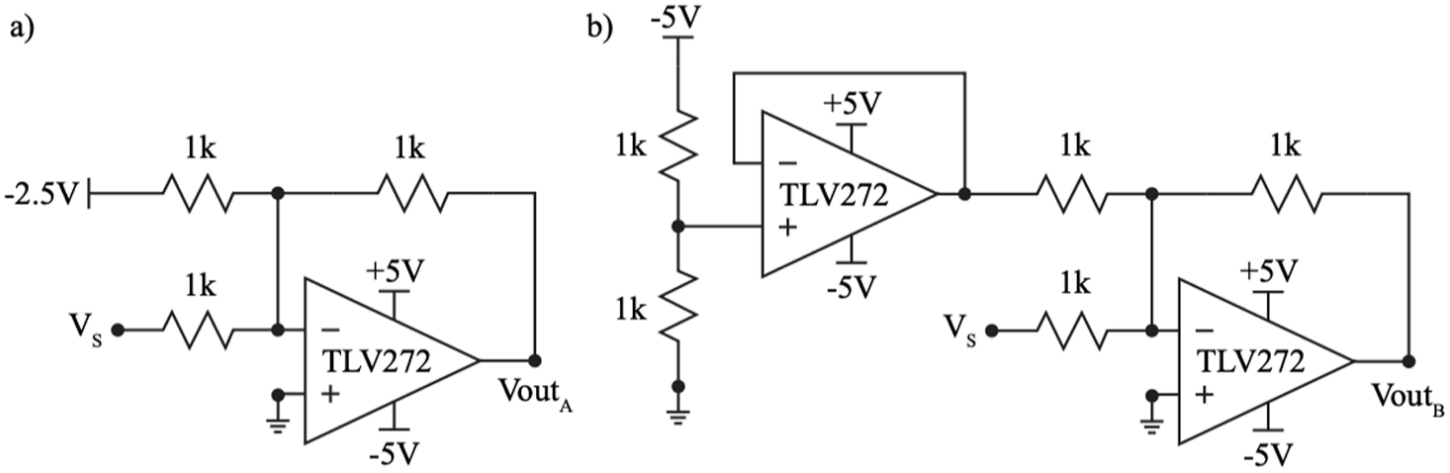
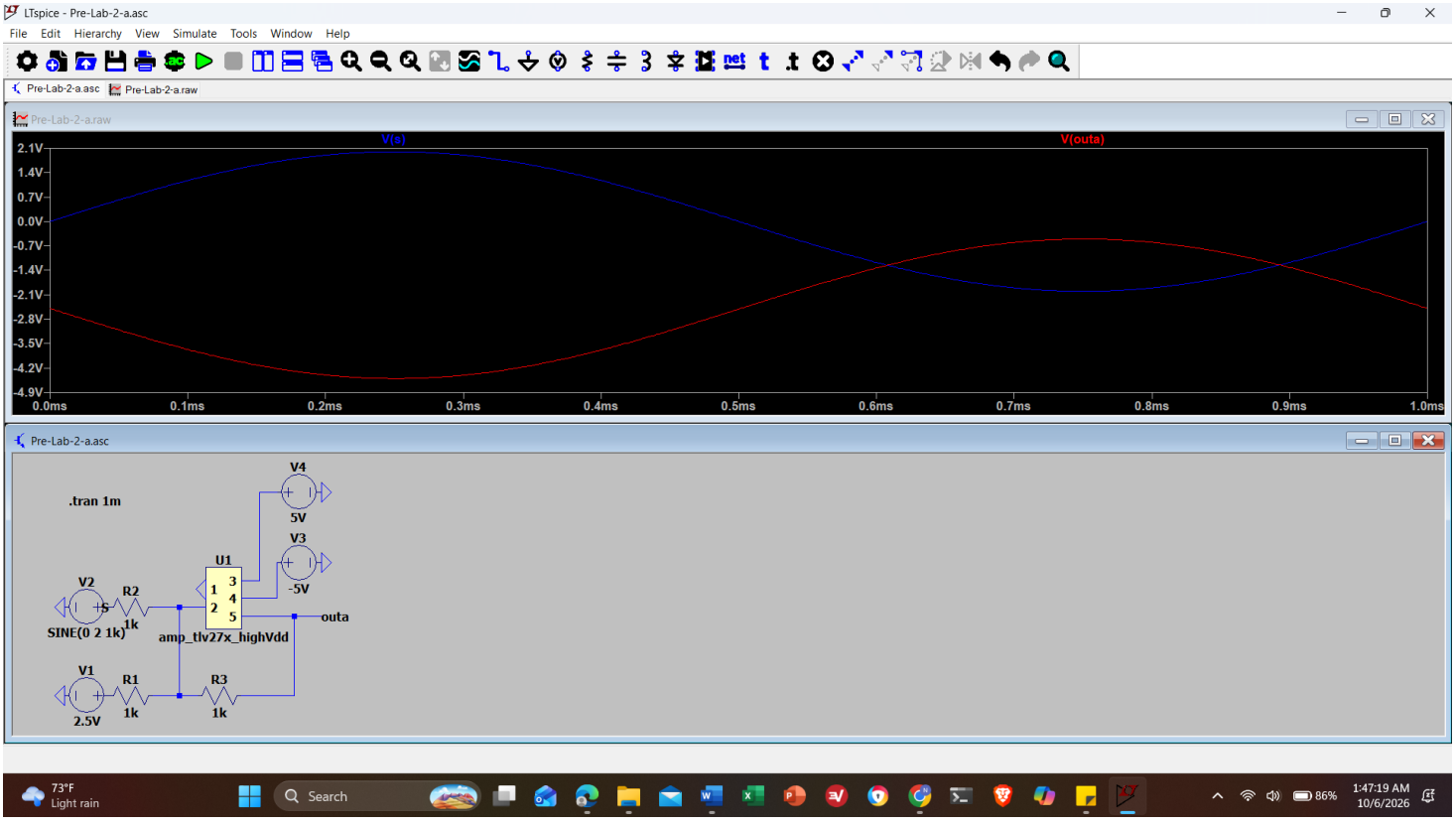


Figure 4: A summing amplifier used to add 2.5V to –VIN (a) and the same configuration where the -2.5V source has been replaced with a buffered voltage divider (b).

1. Simulate the circuit in Figure 6.4 (a) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.

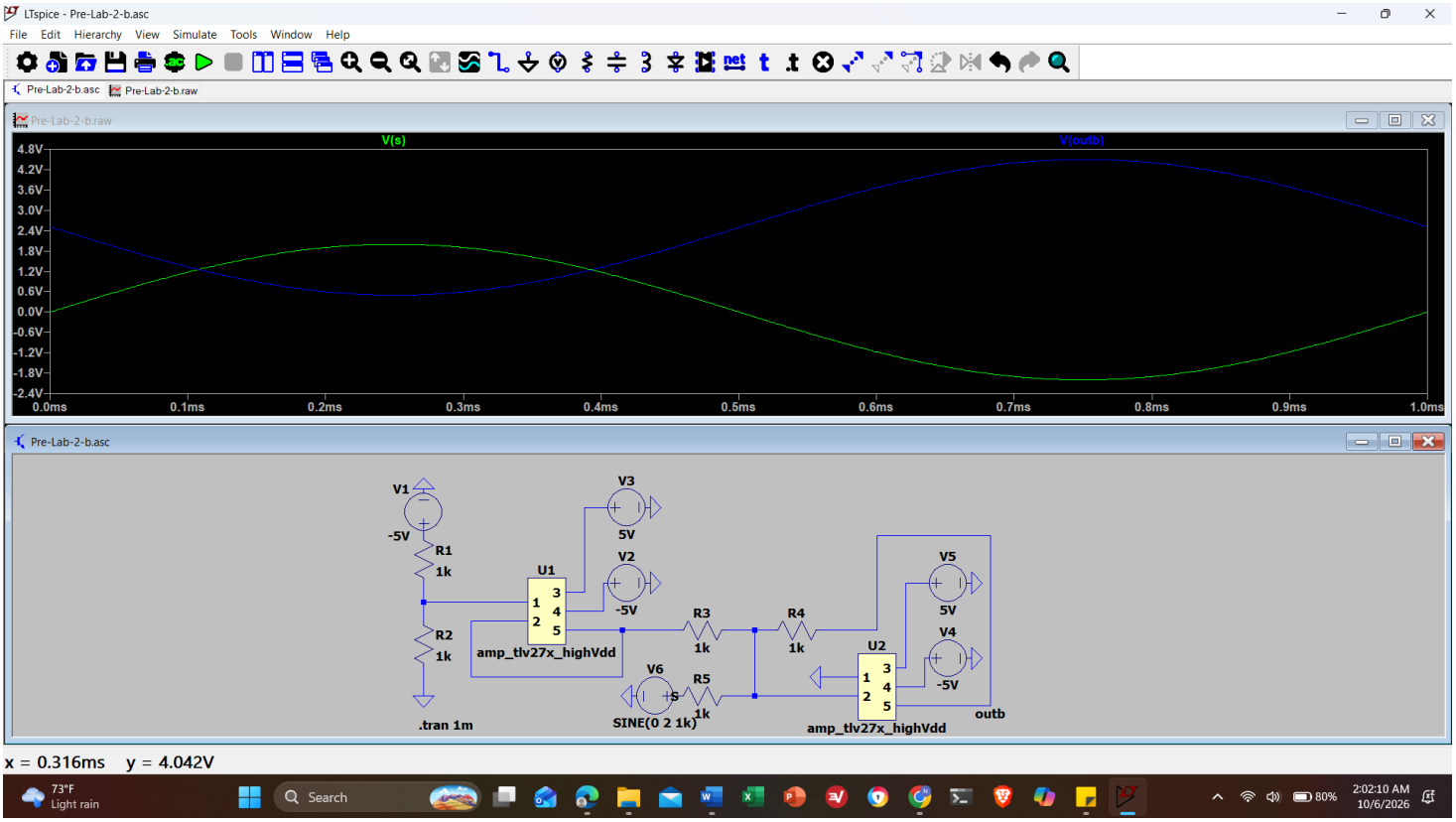


Screenshot 2: Figure 6.4 (a) using a transient analysis with a stop time of 1m (.tran 1m) with plotted input and output voltages

2. Explain why the resistor divider in Figure 6.4 (b) uses -5 V instead of 5 V.

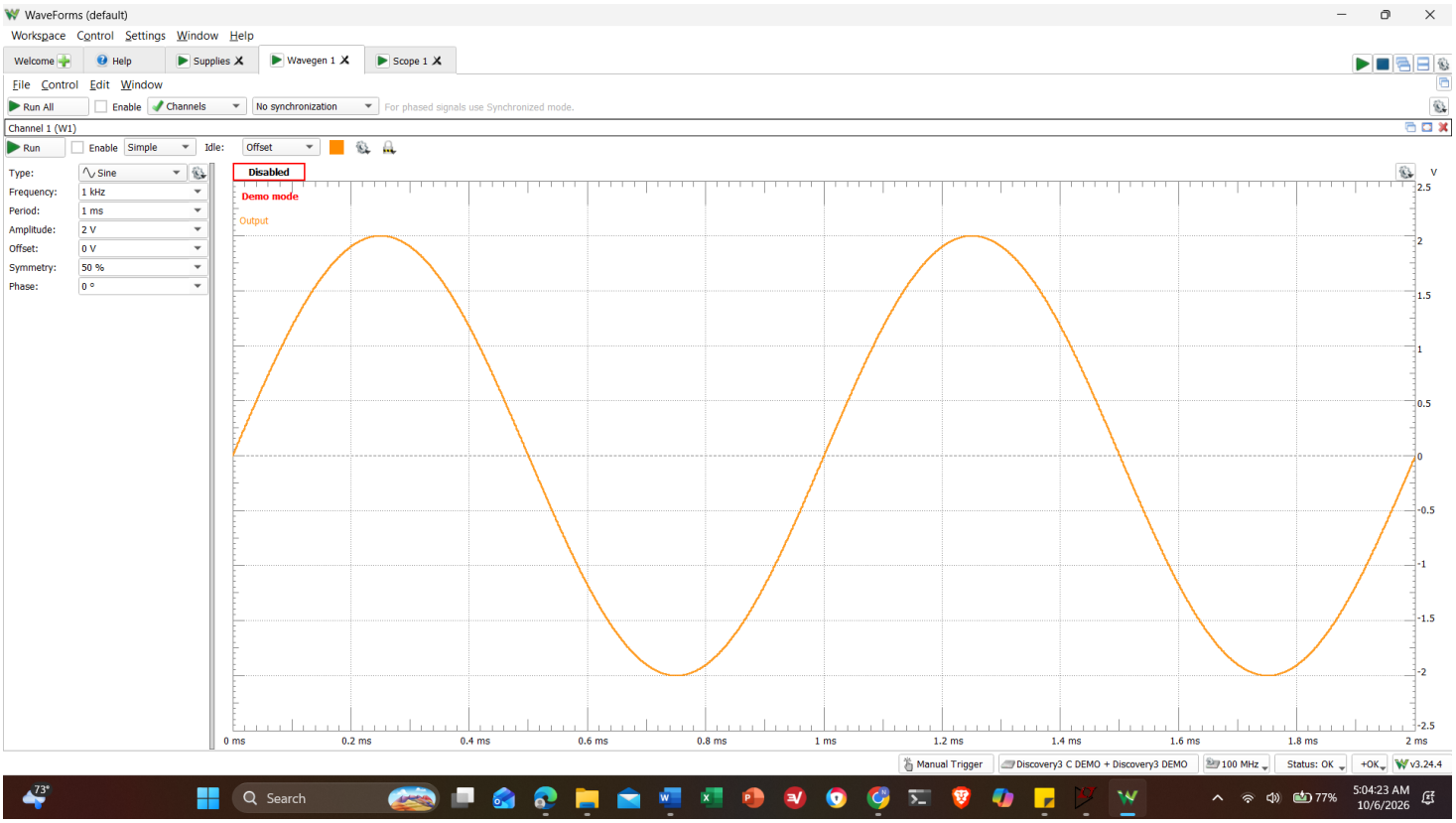
   The resistor divider in Figure 6.4 (b) uses -5V instead of 5V because the circuit is an inverting summing amplifier configuration. In order to have a positive DC shift at the output, a negative reference voltage must be fed into the inverting input.

3. Practically, a 2.5 V reference may not be available for use and instead must be self-generated. Simulate the circuit in Figure 6.4 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.

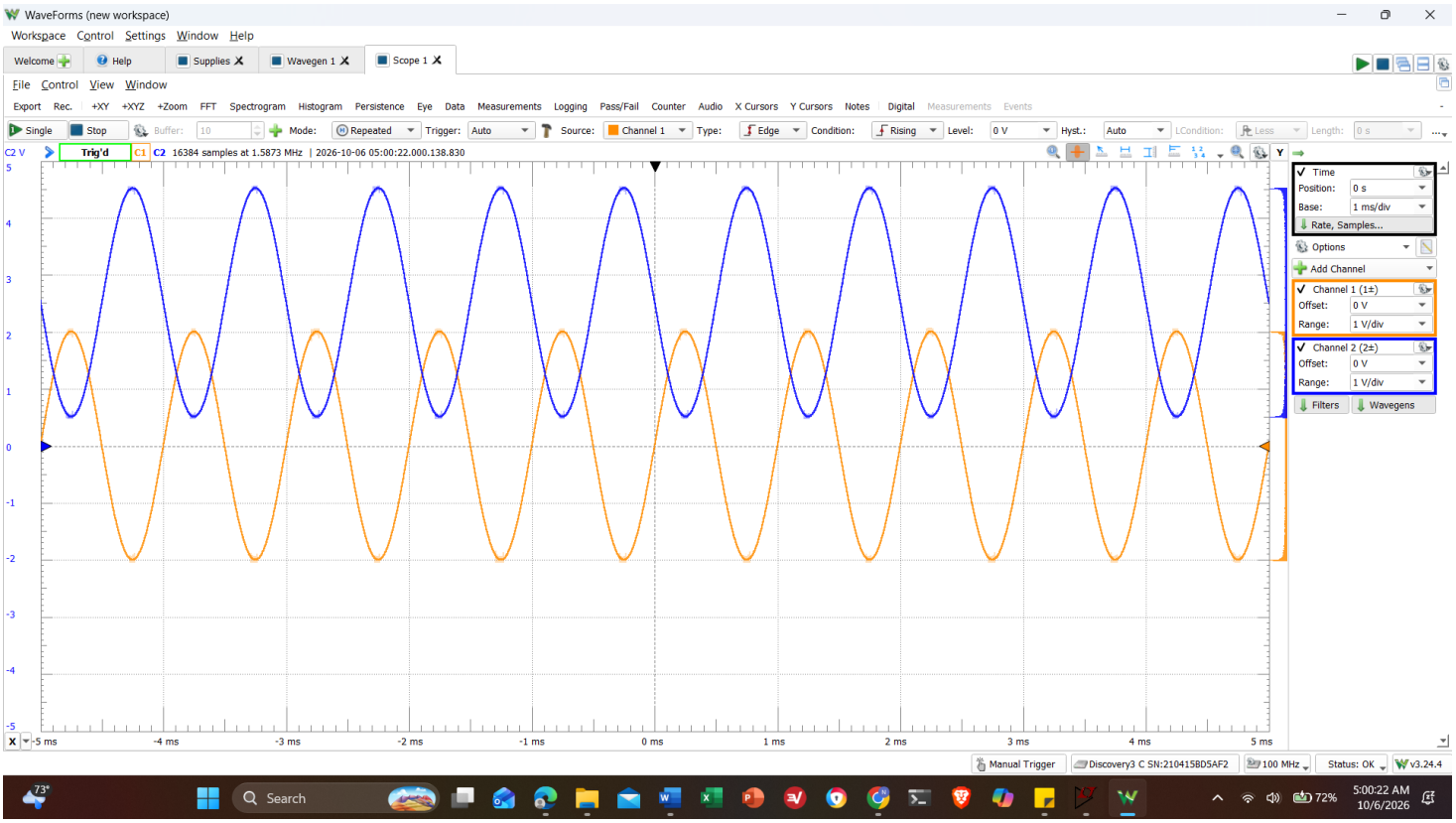


Screenshot 3: Figure 6.4 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency with plotted input and output voltages.

4. Construct the circuit in Figure 6.4 (b) on a breadboard. Configure the Wavegen to generate a sine wave at 1 kHz with a 2 V amplitude and a 0 V offset. Plot the input and output on the Scope and save the circuit to demo at the start of lab. If you see clipping at your output, change the three 1kΩ resistors in the summing op amp to 33kΩ.



Screenshot 4: Figure 6.4 (b) on a breadboard. Configure the Wavegen to generate a sine wave at 1 kHz with a 2 V amplitude and a 0 V offset. Wavegen Window



Screenshot 5: Figure 6.4 (b) on a breadboard. Configure the Wavegen to generate a sine wave at 1 kHz with a 2 V amplitude and a 0 V offset. CH1 is input CH2 is output

5. In the more general case, the full range of input must be accounted for. Simulate the circuit in Figure 6.5 using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN} to a sine wave with a 0 V DC offset, 5 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.

   This circuit uses 3 op amps to convert a +/- 5 V sine wave into a +5/0 V sine wave. The top op amp generates a -2.5 V offset, while the bottom op amp buffers half of the input voltage. Both outputs are then fed into an (inverting) summing op amp to produce V_{outB}. In short, this circuits adds -2.5 V to half of a +/-5 V sine wave, then inverts it to obtain:

$$-\left[\left(\frac{-5}{2}\right)+(-2.5)\right]=5V \; and \; -\left[\left(\frac{5}{2}\right)+(-2.5)\right]=0V$$

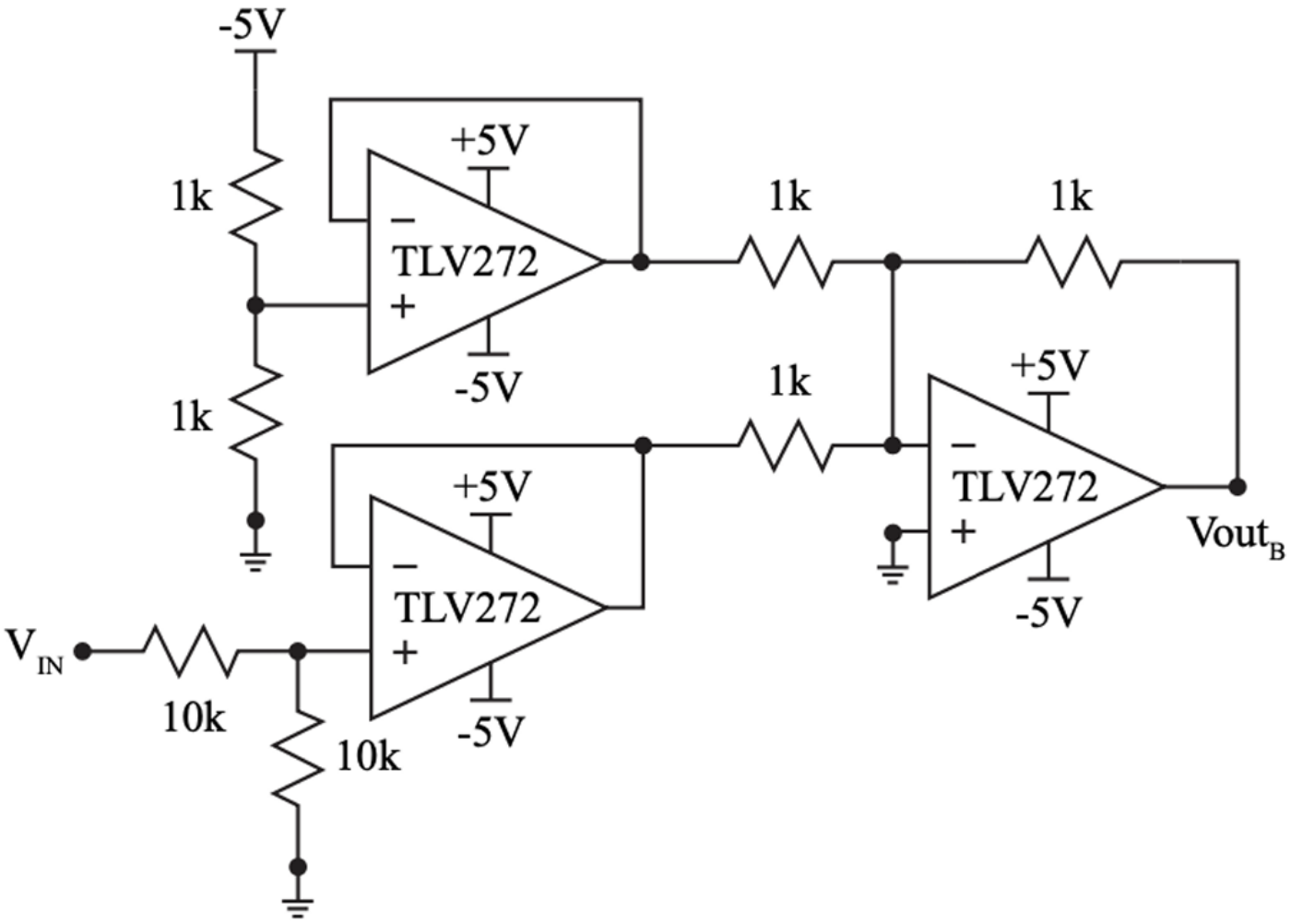
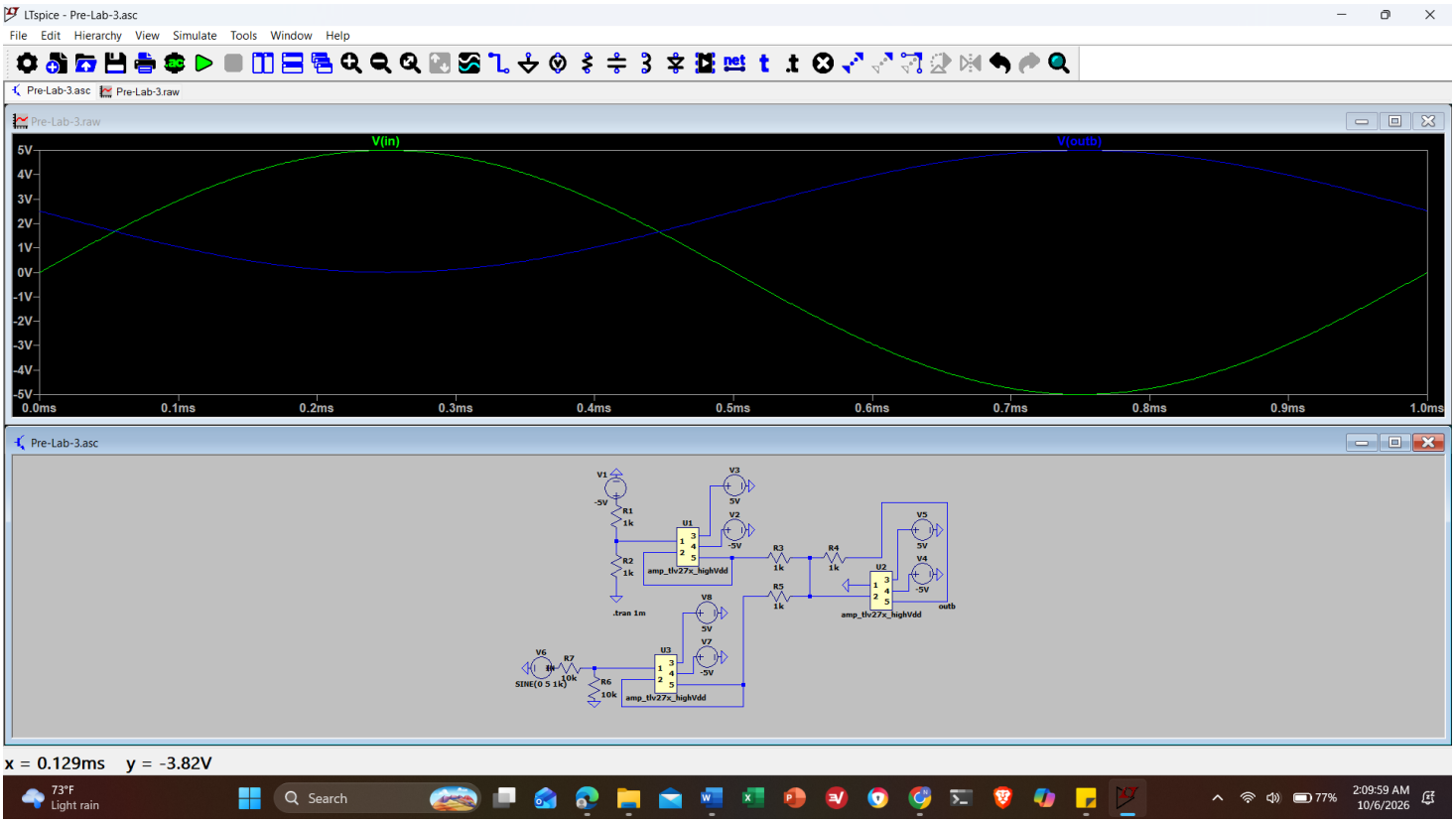


Figure 5: A DC level shifter configured to convert the full range of +/-5 V to +5/0 V.



Screenshot 6: Figure 6.5 using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN} to a sine wave with a 0 V DC offset, 5 V amplitude, and a 1 kHz frequency with plotted input and output voltages

1.5.3 Differential to Single Ended Conversion

Differential signals, which are two signals 180 degrees out of phase, offer improved noise performance because any interference affecting both signals in the same way cancels out when they are combined. In other words, only the difference between the signals is measured, removing unwanted noise. A difference amplifier can be used to convert the differential signal into a single-ended output.

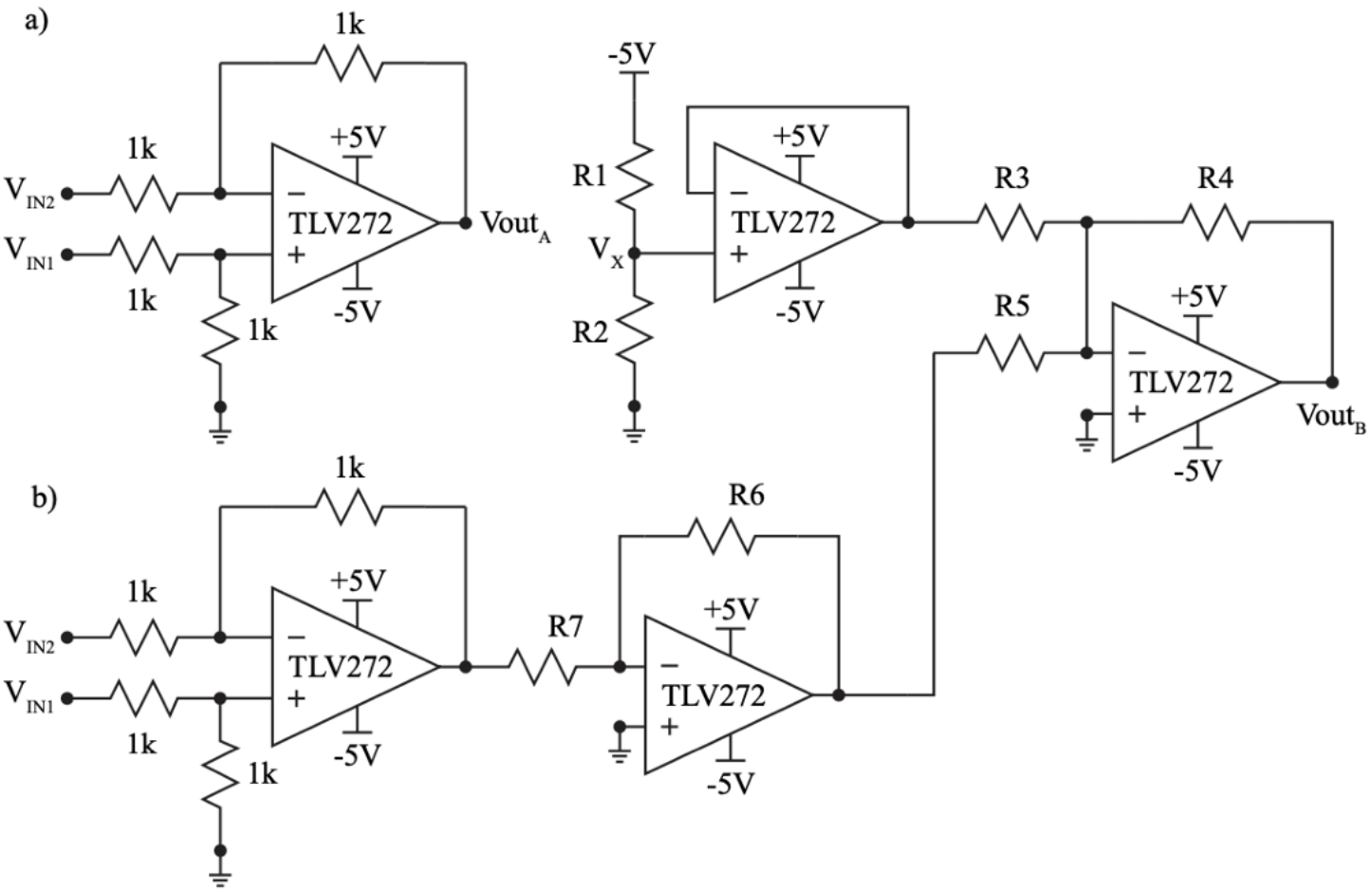
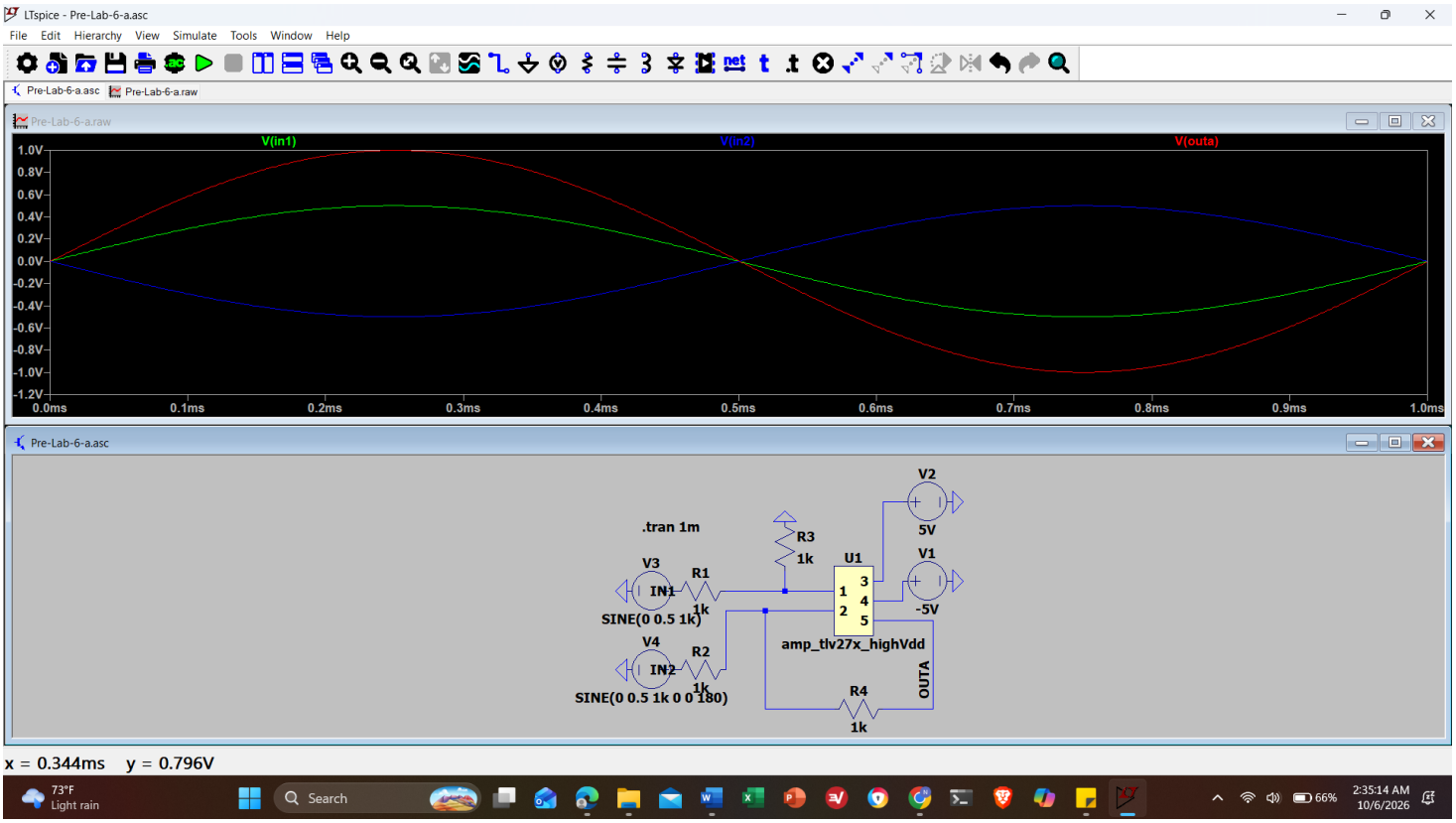


Figure 6: A difference amplifier that converts a differential signal to single ended (a) and a combination of the level shifter and differential to single ended converter (b).

1. Simulate the circuit in Figure 6.6 (a) using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN1} to a sine wave with a 0 V DC offset, 0.5 V amplitude, and a 1k Hz frequency. Set V_{IN2} to a sine wave with a 0 V DC offset, 0.5 V amplitude, a 1k Hz frequency, and a phase of 180 degrees. Save an image of the circuit and a plot of V_{IN1}, V_{IN2}, and V_{outA}.



Screenshot 7

2. Determine the resistor values for R_1 through R_7, recall that there are limited quantities in the lab kit. Choose R_1 and R_2 so that V_X is 2.5 V, R_3, R_4, and R_5 so that both voltages are added equally, and R_6 and R_7 so that the output, V_{outB} utilizes as much of the full range, 0 to 5 V, as possible without clipping. You do not need to use resistors in parallel/series combinations to get an exact value.

   $R_1 = 10k\Omega$, $R_2 = 10k\Omega$, $R_3 = 33k\Omega$, $R_4 = 33k\Omega$, $R_5 = 33k\Omega$, $R_6 = 2.2k\Omega$, $R_7 = 1k\Omega$.

   R_1 and R_2:

$$V_X = -5V\left(\frac{R_2}{R_1 + R_2}\right) => -2.5V = -5V\left(\frac{R_2}{R_1 + R_2}\right) => R_1 = R_2$$

   R_3, R_4, R_5:

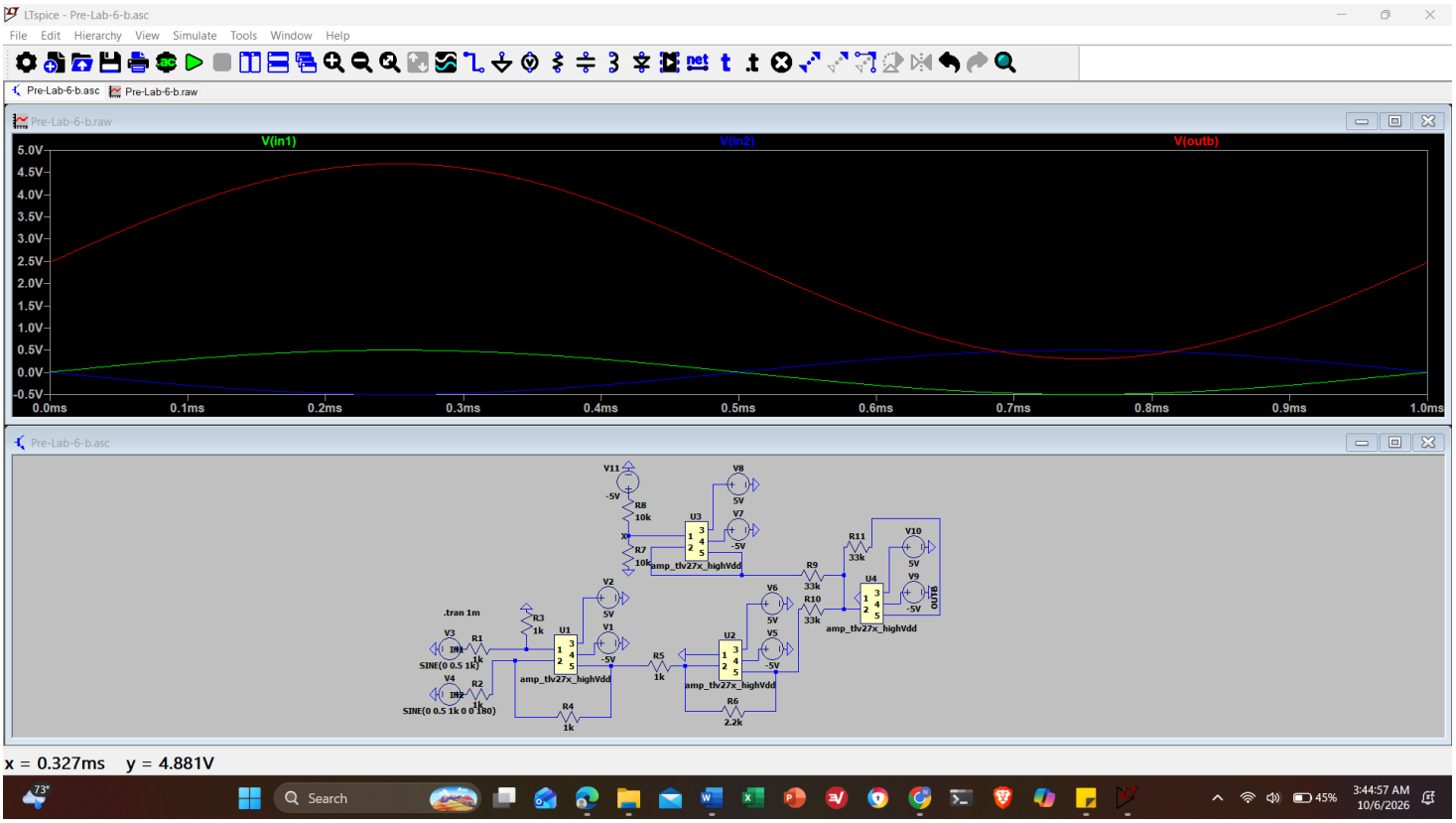
$$V_{out} = -R_f\left(\frac{V_1}{R_{IN1}} + \frac{V_2}{R_{IN2}}\right) => \frac{R_f}{R_{IN}} = 1$$

   Values $R_{3\text{-}5}$ must be greater than $1k\Omega$ to avoid clipping.

   R_6 and R_7:

$$V_{out} = V_{in}\,\frac{R_f}{R_{in}}$$

3. Simulate the circuit in Figure 6.6 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_{S1} and V_{S2} to their previously defined values. Save an image of the circuit and a plot of V_{IN1}, V_{IN2}, and V_{outB}.



Screenshot 8: Figure 6.6 (b) using a transient analysis with a stop time of 1m (.tran 1m) with inputs and output plotted

4. Explain how the circuit in Figure 6.6 (b) works in a few sentences, discussing the purpose of each op amp.

   The first op-amp in the bottom left takes the two different input signals and outputs the voltage difference between them to convert the differential signal into a single-ended signal while rejecting noise. The next op-amp in the bottom middle takes the single-ended output from the difference amplifier and applies a gain based on the R_6 to R_7 ratio to amplify the AC signal amplitude so the output has a 0V-5V range without clipping. The next op-amp in the top middle is a voltage buffer configured as a voltage follower that is used to isolate the voltage divider network (R_1 and R_2) from the rest of the circuit to provide a stable, high-current DC reference voltage (V_X) derived from a -5V supply without loading effects. The final op-amp on the right is an inverting summing amplifier that combines the amplified AC signal from the inverting amplifier and steady DC offset from the voltage buffer to add them together and invert them to produce the final level-shifted output that operates cleanly within the voltage bounds.

1.5.4 Pre-Lab Checklist

The following must be completed at the start of lab.

1. Section 6.5.1
   a. Item 1: Image of the circuit and a table of output voltages.
   b. Item 2: Answer to question.

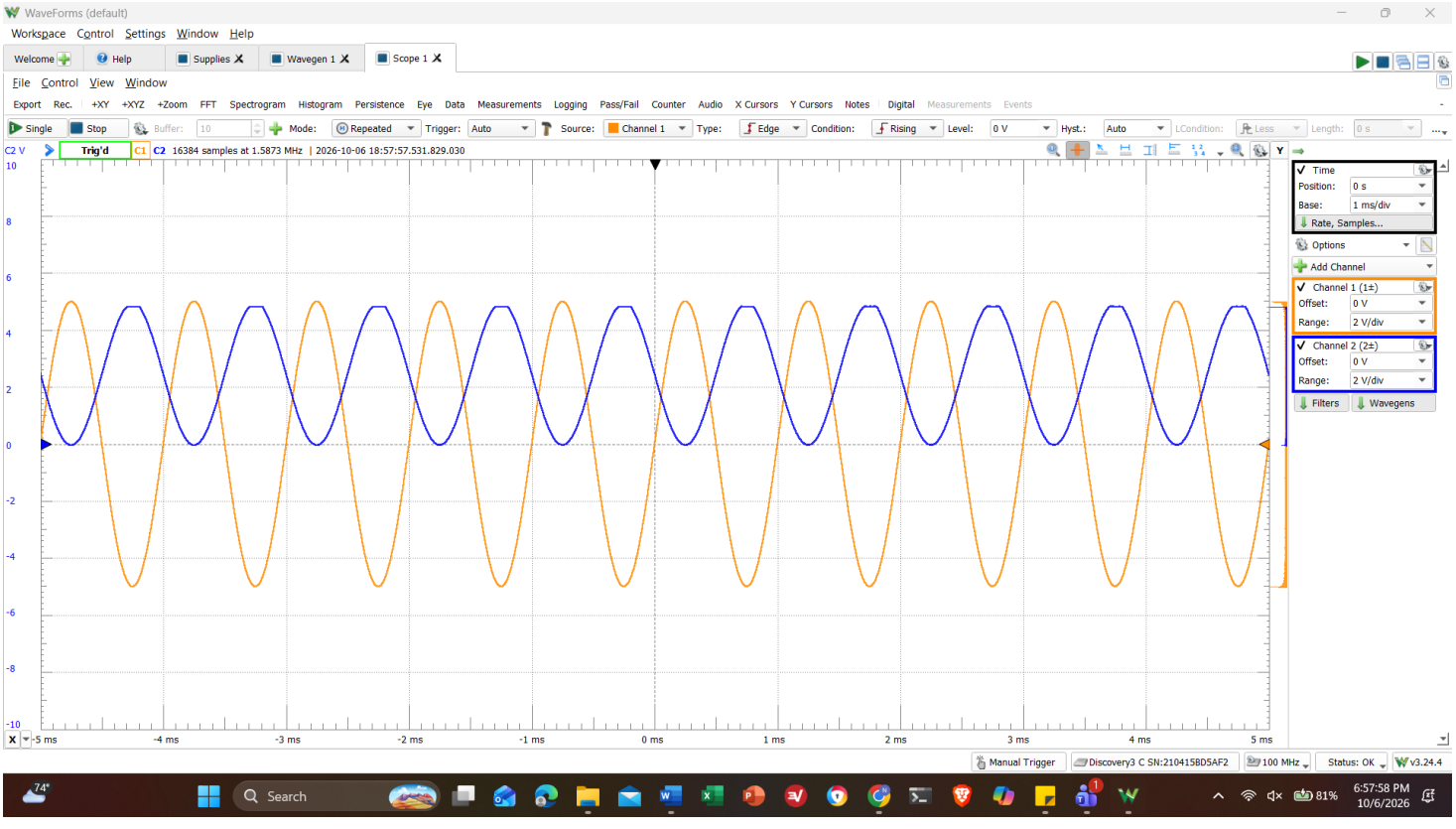
2. Section 6.5.2
    a. Item 1: Image of the circuit and a plot of the input and output voltages.
    b. Item 2: Answer to question.
    c. Item 3: Image of the circuit and a plot of the input and output voltages.
    d. Item 4: Plot of the input and output voltages and a circuit demonstration.
    e. Item 5: Image of the circuit and a plot of the input and output voltages.
3. Section 6.5.3
    a. Item 1: Image of the circuit and a plot of the input and output voltages.
    b. Item 3: Image of the circuit and a plot of the input and output voltages.
    c. Item 4: Answer to question.

1.6 In-Lab Requirements

Complete the following in lab.

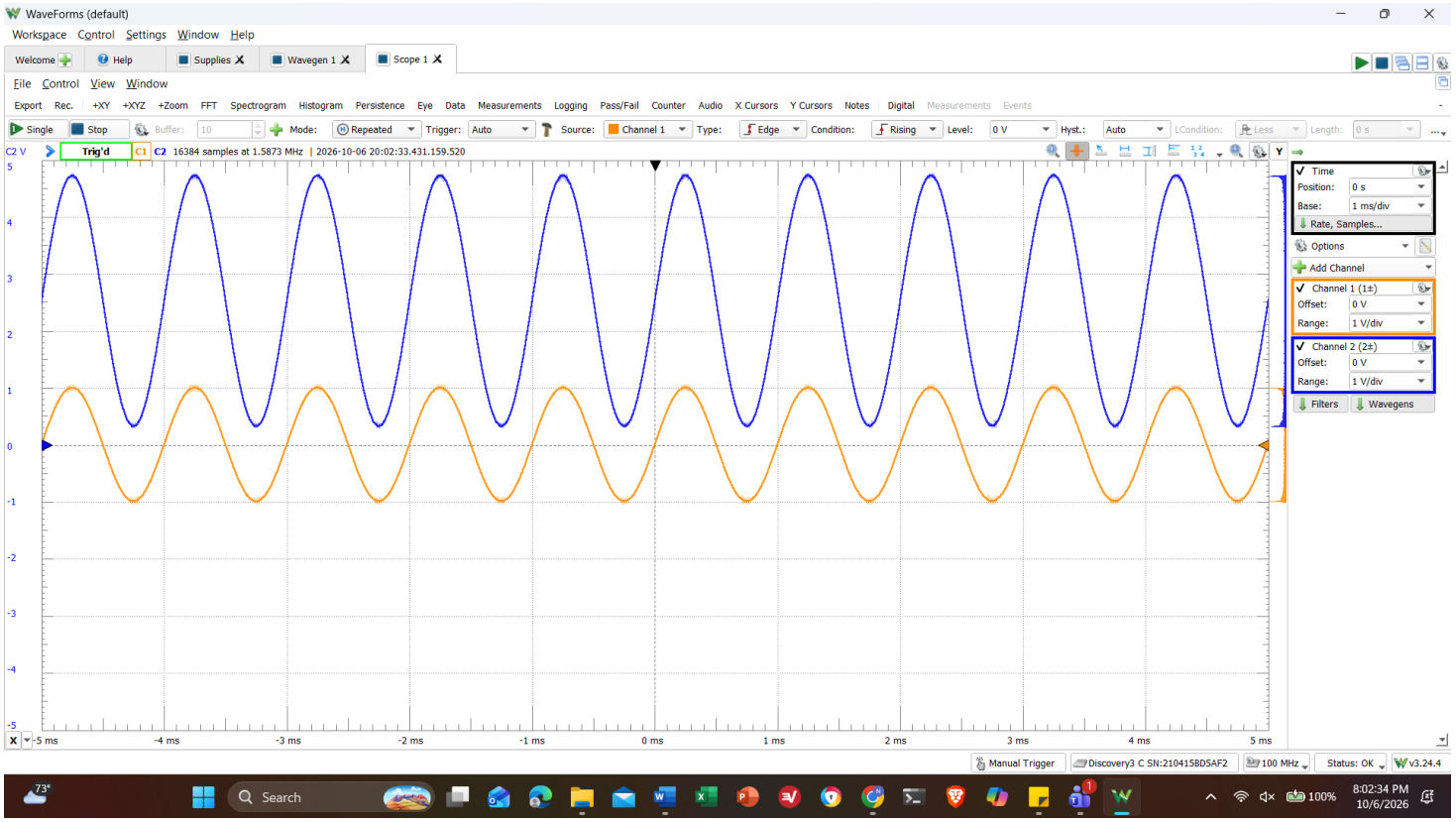
1.6.1 Construction

1. Construct the circuit from Section 6.5.2 Item 5. Set Wavegen 1, V_{IN} to a sine wave with a 5 V amplitude, 0 V offset, and a frequency of 1 kHz.
2. Plot the input and output voltages, save an image of the plot.



Screenshot 9: Circuit from Section 6.5.2 Item 5. Set Wavegen 1, VIN to a sine wave with a 5 V amplitude, 0 V offset, and a frequency of 1 kHz. CH1 is input, CH2 is output

3. Construct the circuit from Section 6.5.3 Item 3. Set Wavegen 1 to V_{IN1} and Wavegen 2 to V_{IN2}. Synchronize Wavegen 1 and Wavegen 2 using the dropdown in the wavegen tab of waveforms. It is located two tabs to the right of "Run all" and is defaulted to "No synchronization".
4. Plot the input (differential) and the output voltages, then save an image of the plot. Typically, scope wire 2 (with the white stripe) is connected to ground because scopes measure voltages relative to another signal. In this case, however, you connect scope wire 1 to Wavegen 1 and scope wire 2 to Wavegen 2, allowing the scope to measure the difference between them.



Screenshot 10: Circuit from Section 6.5.3 Item 3. Set Wavegen 1 to VIN1 and Wavegen 2 to VIN2. Synchronize Wavegen 1 and Wavegen 2 using the dropdown in the wavegen tab of waveforms. CH1 is input and CH2 is output.

1.7 Write Up

Include the following in the write up.

1. Section 6.6.1
   a. Item 1: LTspice schematic of the circuit.
   b. Item 2: Plot of the input and output voltages.
   c. Item 3: LTspice schematic of the circuit with all calculated resistors.
   d. Item 4: Plot of the input and output voltages.

Discussion Question: Discuss the two circuits constructed in lab, DC level shifter and differential to single ended converter. Explain how each circuit works at the block level and how

resistor tolerance could or could not influence the output. Also explain the design choices for the circuit, Section 6.5.3 Item 3, constructed in lab. Which resistors were chosen and why?

The circuit in Figure 6.6 (b) works in two stages. The first op-amp acts as a difference amplifier, measuring the voltage difference between the differential inputs to remove common-mode noise and outputting a single-ended bi-polar signal. The second and third op-amps form a DC level shifter, where one buffers the negative voltage reference and the other acts as an inverting summing amplifier to shift the single-ended bi-polar signal to a strictly positive 0V-5V range suitable for digital processing. Resistor tolerances in these stages will impact the common-mode rejection ratio and the accuracy of the DC offset, potentially leading to clipped signals or residual noise at the output.

For Section 6.5.3 Item 3, I used $R_1 = 10k\Omega$, $R_2 = 10k\Omega$, $R_3 = 33k\Omega$, $R_4 = 33k\Omega$, $R_5 = 33k\Omega$, $R_6 = 2.2k\Omega$, $R_7 = 1k\Omega$ for the resistors. R1 = 10kΩ and R2 = 10kΩ because they form a voltage divider between the -5V supply rail and ground to generate a stable DC reference voltage; using the voltage divider formula, R1=R2 splits the voltage in half, and 10kΩ parts were available in the lab kit. R3, R4, and R5 configure the final op-amp as an inverting summing amplifier. To add both inputs equally without adding extra signal gain, the ratio of the feedback resistor (R_4) to each input resistor (R_3 and R_5) must be equal to 1 since $R_f/R_{in} = 1$. The 33kΩ was chosen because it fit with the kit that we were given, and it was used instead of 1kΩ to prevent clipping. R_7 and R_6 were the resistors that set the gain of the inverting amplifier stage that processes the differential output. The differential input signals have an amplitude of 0.5V and are 180 degrees out of phase, which creates a 1.0V peak-to-peak AC swing. To scale the signal so that it utilizes as much of the 0V-5V range as possible without rail-to-rail clipping, a target gain of 2.5 is needed so $R_7 = 1k\Omega$ and $R_6 = 2.2k\Omega$ works since gain is R_6/R_7 and would be the closest based on what we have in the toolkit.

CONCLUSION

This lab demonstrated the application of operational amplifiers for signal conditioning tasks. By simulating and constructing buffer circuits, level shifters, and difference amplifiers, the ability to isolate circuit stages, manipulate DC offsets, and extract differential signals while rejecting common-mode noise was validated.